INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
End-Spring Semester Examination 2023-24
Date of Examination: 22.04.24 Session: AN Duration: 3 hrs Full Marks: 60
Subject No.: AI60004 Subject: Big Data processing
Dept/Center/School: Centre of Excellence in Artificial Intelligence
Specific charts, graph paper, log book required: NO

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1. *Consider the following two paired RDDs,* P = {(1,2), (1,6), (2,5), (3,6)} and Q = {(1,8), (2,6)}. What *would be the content of OUT in each of the following code snippets?* **(4 x 2.5)**

a) T = P.join(Q)
OUT = T.cogroup(Q)

b) R = P.join(Q)
OUT = R.groupByKey()

c) OUT = P.mapValues(x => x+1).reduceBykey(x,y => x+y)

d) T = P. cogroup(Q)
OUT = T.lookup(1)

(5 x 2)

2. a) What *precaution needs* to be taken before the following operation is performed on an RDD R R.groupByKey().*map(row* => (row._1, row._2.reduce(addFunc))).collect()? Assume that there is no *syntax error* in any part of the code and addFunc() only adds values for the same key.

b) Which of the operations from among reduceByKey(), combineByKey(), and lookup() benefit from partitioning? Justify your answer.

c) In an iterative spark code, it performs five RDD transformation operations in each iteration. The result of the last operation will be required in the next iteration. What is a good way to do it? Choose the best option. Assume that the final RDD size is very small and every node can hold it in its memory. No marks will be given without justification.

i) Use collect() function
ii) Persist the last result in memory as an RDD and then use lookup
iii) Broadcast the RDD
iv) Broadcast the RDD and cache it.

d) What are the relationships (in terms of type and structure) between key and value pairs of mapper, combiner and reducer in map-reduce program?

e) Compute the time complexity of finding only the heavy hitter triangles (where each vertex of the triangle is a heavy hitter node).

(5 + 5)

3. a) What are the main steps of processing a *resilient distributed dataset* (RDD)? For the following code snippet, associate each of the spark operations to the corresponding step that you mentioned (write this in a two-column table where the first column is the spark operation and second column is the step no.)

```
data = sc.textFile("input-data.txt")
pairs = data.flatMap(lambda line: line.split(" ")).map(lambda word: (word, 1)).reduceByKey(lambda a, b: a + b).cache()
filteredWords = pairs.filter(lambda x: x[1] > 2)
filteredWords.saveAsTextFile("outputFile")
```

b) You are given a set of n English sentences each having 100 bytes of data. In order to compute the similarity between every pair of sentences, a programmer writes a map-reduce code where the key of the mapper is pairs of sentence ids (sid1, sid2) where (sid1 < sid2) and the value is the text of the any one of the sentences of (sid1, sid2). Assuming that a sentence id takes 4 bytes of data, what is the amount of data being generated by the mapper? Suggest an approach to reduce data communication for this problem.

(6 + 4)

4. ~~a)~~ You are given a graph with n vertices and your goal is to find minimum cut partition (in two groups) using Karger's randomized algorithm. Derive how many times you need to run Karger's algorithm with independent random choices so that the highest probability of not finding a minimum cut is $\frac{1}{n}$. (No marks will be given without derivation).

~~b)~~ Consider an unweighted undirected graph G with n vertices and m edges. Girvan-Newman algorithm in its very first iteration produces k partition of the graph, where the betweenness centrality values of each of the edges that it removes are the same. Compute the maximum number of triangle that the graph G might have under any possible edge configuration.

(4 + 6)

5. a) Illustrate with examples the working mechanism of groupByKey() and reduceByKey() operations in Spark.

b) In map-reduce, how does value-to-key design pattern enhance scalability? What is the role of custom partitioner in value-to-key design pattern? Design a custom partitioner in value-to-key design pattern for creating inverted index in search engines.

(6 + 4)

6. a) What are the output modes in structured spark streaming? Provide examples for each case.

b) In spark, what do the following operations take as input and what do they produce?

i) *reduceByKeyAndWindow* and ii) *countByValueAndWindow*